Hasan Emre USTA

(+90) 506 691 09 80 | hasanemreusta5@gmail.com | linkedin.com/in/hasanemreusta/ | github.com/hasanemreusta

EDUCATION

Cankaya University, Ankara - *B.Sc. in Computer Engineering* | Sep 2021 - Aug 2026

EXPERIENCE

Software Engineering Intern – Backend Team- Ulusal Factoring | Feb 2026 - Present

- Built a device–user match approval feature on the core factoring platform using **C#, .NET, and PostgreSQL**, with **NUnit** unit tests.
- Provided ongoing support for the platform, including debugging, resolving day-to-day issues, and small feature and data-quality improvements.
- Improved a **Python/Tesseract OCR** document-processing service, significantly reducing processing time and raising recognition accuracy.
- Deployed a **machine learning-based scoring model** as a production **FastAPI** service and integrated it into the company's existing .NET infrastructure.

Summer Internship - Finagotech (Payment Systems Department) | Jun 2024 - Jul 2024

- Developed a **RESTful API** in **C++** with **SQL Server** integration and implemented secure CRUD operations for customer data.
- Built **a C# WPF** desktop client for API interaction and analyzed POS transaction authorization flow within the SDLC.

PROJECTS

Kim Oynadı - Real-Time Multiplayer Football Quiz Game - *Personal Project | 2026*

- Built and launched a 1v1 mobile quiz where players race to name a footballer who played for both of two server-chosen clubs, using **Flutter/Riverpod, Node.js/TypeScript, Socket.IO, Redis, and PostgreSQL** in an npm-workspaces monorepo with shared TypeScript event types.
- Designed a **server-authoritative round state machine** (TTL'd Redis state, reconnect handling, matchmaking queue, Elo-based ranked mode) with 24,000+ team-pair common-player lists precomputed, so no live queries run during a round.
- Deployed on a Hetzner VPS behind Cloudflare with **Docker and GitHub Actions CI/CD**; load-tested to 8,000 concurrent bot connections with no dropped connections (ack p95 under 92 ms); submitted to the App Store and in Google Play closed testing.

Mystery Shorts Bot - Multi-Agent Content Pipeline - *Personal Project | 2026*

- Built an autonomous **7-stage multi-agent pipeline** orchestrated via **Claude Code** that researches, writes, voices, illustrates, edits, and publishes short-form videos end to end using **Python, Whisper, MoviePy, and the YouTube Data API**.
- Designed a deterministic, idempotent pipeline with a **QA gate** between generation and rendering, per-scene retry/fallback logic, and a human-in-the-loop review step before publishing;

IoT-Integrated Accessible Smart Home System - *CENG 407-408 (Senior Project) | 2025-2026*

- Developed an accessibility-focused smart home platform enabling hands-free control via eye movements and blinks, using **Raspberry Pi 5, OpenCV, MediaPipe, FastAPI, React Native, and python-kasa**.
- Implemented event-driven architecture with local edge vision processing (gaze + blink detection) feeding a modular **FastAPI** backend and a **React Native** mobile app; end-to-end demo controlling a TP-Link/Tapo smart bulb in real time. Wiki: github.com/CankayaUniversity/ceng-407-408-2025-2026-IoT-Integrated-Accessible-Smart-Home-System/wiki

Turkish Legal Q&A RAG Pipeline - *CENG 493 (Natural Language Processing) | May 2026*

- Built an end-to-end **Retrieval-Augmented Generation (RAG)** system over 41,973 Turkish legal articles using **Python, HuggingFace, FAISS, BM25, and QLoRA**; fine-tuned embedding, reranker, and LLM components via 8-cell ablation study.
- Achieved **7.8x Citation F1 improvement** and +32.3 pp Recall@10 over the BM25 baseline; GitHub: github.com/hasanemreusta/turkish-legal-rag

Online Retail Big Data Analytics & Recommendation System - *CENG 476 (Big Data) | May 2025*

- Engineered an end-to-end real-time retail analytics pipeline using **Apache Spark, Kafka, MongoDB, Flask, and PySpark ML**, processing streaming data with sub-second latency.
- Delivered an **ALS-based recommendation engine** and a live **Flask** dashboard visualizing pipeline metrics.

Classroom Reservation System - *CENG 382 (Web Development) | May 2025*

- Developed an **ASP.NET Core MVC** application with **EF Core and SQL Server** for classroom reservation management.
- Implemented **JWT authentication**, role-based access control, and automated conflict detection to eliminate double-booking errors.

ACHIEVEMENTS & CERTIFICATIONS

- Learning SQL with Applications - BTK Online Course Certificate (Jul 2024).
- Introduction to **Model Context Protocol** - **Anthropic** Certificate (May 2026). Verify: https://verify.skilljar.com/c/wncimb83hf8w
- Introduction to **Agent Skills** - **Anthropic** Certificate (Jun 2026). Verify: https://verify.skilljar.com/c/kupm5s7ymz2a
- Ranked in the top 2% among ~3 million candidates in Turkiye's 2021 Higher Education Entrance Exam (YKS).
- Completed the 40-hour Mechatronics Workshop by the Ministry of National Education and participated in TUBITAK 4006 Science Fair.